

BOARDROOM-READY MATH · 2026

Math Layouts

Seven variants under one `math` class. For quants, mathematicians, physicists, ML researchers, statisticians, and economists. KaTeX renders `$...$` inline and `$$...$$` display; the layouts surround the equations with the structure each audience expects.

SECTION 01

The hero equation · bare **math**
resolves to feature.

The closed-form estimator.

$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

WHERE

- $\hat{\beta}$ — OLS coefficient vector
- X — design matrix, $n \times p$
- y — response vector, length n
- $X^{\top} X$ — Gram matrix, $p \times p$, must be invertible

SECTION 02

The derivation chain ·

math.derivation

Derivative of f from first principles.

$f(x + h) = f(x) + f'(x) h + O(h^2)$	<i>Taylor expansion, $n = 2$</i>
$f(x + h) - f(x) = f'(x) h + O(h^2)$	<i>subtract $f(x)$</i>
$\frac{f(x + h) - f(x)}{h} = f'(x) + O(h)$	<i>divide by $h \neq 0$</i>
$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = f'(x)$	take the limit

SECTION 03

Definition · theorem · proof ·
math.theorem

Intermediate Value Theorem.

DEFINITION. A function $f : [a, b] \rightarrow \mathbb{R}$ is *continuous* on $[a, b]$ if $\lim_{x \rightarrow c} f(x) = f(c)$ for every $c \in [a, b]$.

THEOREM. Let f be continuous on $[a, b]$ and let y lie strictly between $f(a)$ and $f(b)$. Then there exists $c \in (a, b)$ with $f(c) = y$.

PROOF. Set $S = \{x \in [a, b] : f(x) < y\}$. S is non-empty and bounded; let $c = \sup S$. Continuity at c forces $f(c) = y$. $\square$

SECTION 04

Side-by-side · `math.compare`

Frequentist vs Bayesian point estimate.

FREQUENTIST

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(y \mid \theta)$$

Maximises the likelihood — no prior. Uncertainty quantified by the sampling distribution of $\hat{\theta}$ across hypothetical repeats.

BAYESIAN

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid y)$$

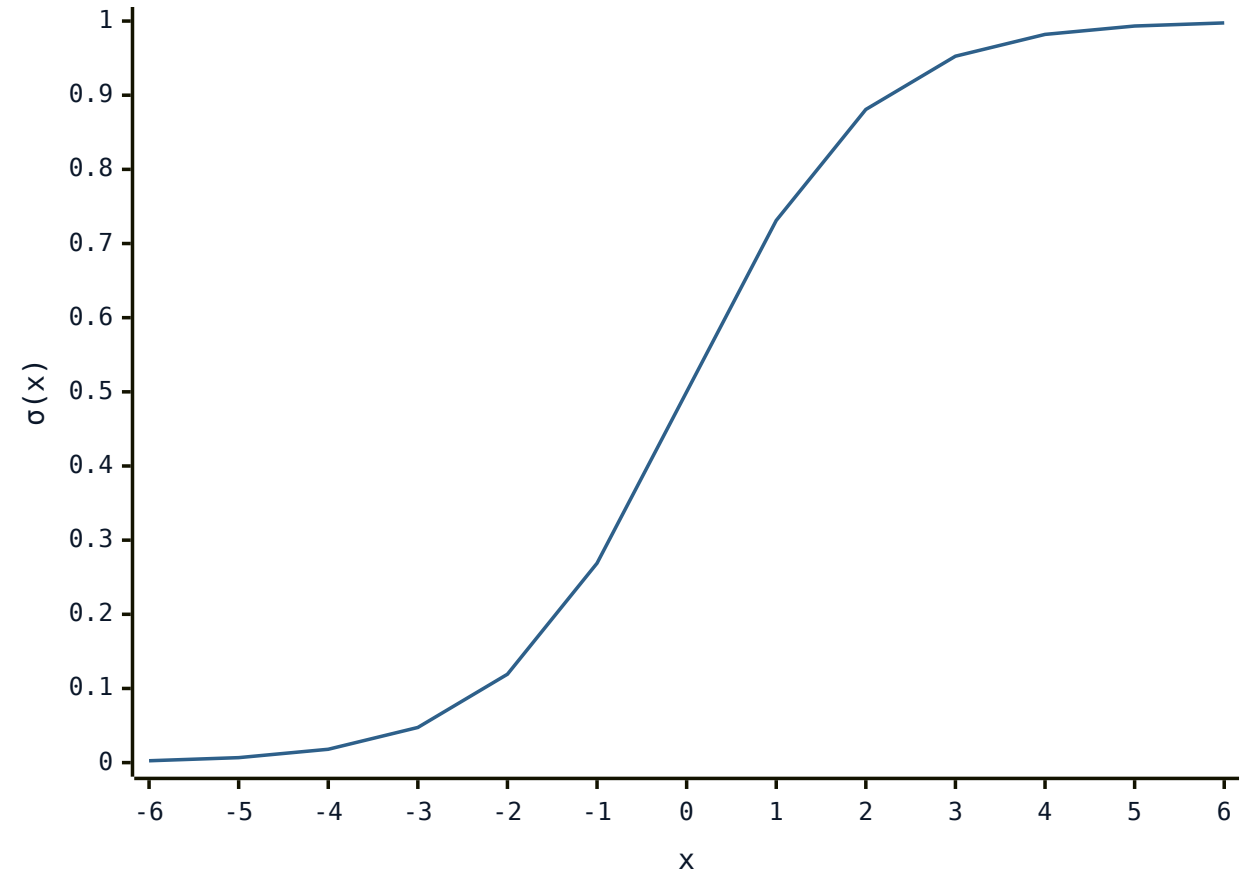
Maximises the posterior — conditions on the prior $p(\theta)$. Uncertainty is the posterior itself, no repeated sampling required.

SECTION 05

Equation + plot · `math.canvas`

The sigmoid.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



Maps $\mathbb{R} \rightarrow (0, 1)$. *S*-shaped, $\sigma(0) = 0.5$, steepest slope at the origin.

SECTION 06

Matrix · `math.matrix` and `math.matrix.decompose`

The design matrix X .

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

- `SHAPE` — $n \times (p + 1)$
- `ROWS` — observations
- `COLS` — intercept + p features
- `RANK` — full-rank for OLS to have a unique solution
- `COLUMN 0` — all-ones, absorbs the intercept

Singular value decomposition.

$$A = U \Sigma V^{\top}$$

COMPONENT MATRICES

$$U \in \mathbb{R}^{m \times m}$$

$$\Sigma \in \mathbb{R}^{m \times n}$$

$$V \in \mathbb{R}^{n \times n}$$

SECTION 07

Statistical result · `math.stats`

Effect of the treatment.

$$\hat{\beta} = 0.42 \pm 0.03$$

95% CI: [0.36, 0.48]

$p < 0.001$ · $n = 1,204$

*For every additional unit of exposure, the outcome rises by 0.42 SD — roughly an **8%** shift on the baseline. Effect size is the headline; the p -value just rules out chance.*

Seven variants. One contract.

*math · math.derivation · math.theorem · math.compare ·
math.canvas · math.matrix · math.stats*

*KaTeX renders the equations; Lattice provides the room around
them.*